

Adrian Bedregal

Gameplay & Engine Programmer | C++ · Unreal Engine 4/5 · Game Systems Architecture

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Open to: Remote (Worldwide) · Relocation Negotiable · Immediately Available

SUMMARY

Gameplay & engine programmer with 6+ years of game development experience (since 2020) — including 3.5+ years in professional industry roles — and 12+ years writing C++. Shipped commercial AAA and AA titles in collaboration with Epic Games and other major studios using Unreal Engine 4/5, C++, and custom engine architecture. Specialised in engine systems, gameplay programming, AI systems, and editor tooling.

EXPERIENCE

Applebomb Games — Remote (Arequipa, Peru)

Lead Programmer (Full-Time since Feb 2026; Part-Time May 2025 – Feb 2026) | May 2025 – Present

Mark The Headless Chicken · C++ | Unreal Engine 5

- Led a team of 10 engineers to deliver core gameplay systems and production features.
- Created AI systems architecture, AI development tools, and Behaviour Trees for core AI-driven features.
- Optimised gameplay and physics systems to maintain stable 60 FPS frame budgets on target platforms.
- Designed architecture for core gameplay systems, UI, physics systems, and game loop integration.

Globant — Remote (Arequipa, Peru)

Software Engineer | May 2024 – Feb 2026

1 AAA Shipped Title & 1 AAA Unannounced Title · C++ | Unreal Engine 5

- Redesigned UI architecture for two core features in Fortnite in collaboration with Epic Games.
- Modified Unreal Engine 5 source code to optimise networking, rendering, and UI subsystems under real-time performance constraints.
- Designed system architecture and built 4 editor tools supporting quest systems, AI workflows, and tutorial pipelines, including backend sync and in-game gameplay elements for an unannounced massive multiplayer title developed for a Japanese publisher (under NDA).
- Contributed to the early architecture and storage design of the inventory system, integrating AI systems and UI widgets.

Bamtang Games — Remote (Arequipa, Peru)

Software Engineer | Oct 2022 – Apr 2024

3 Shipped Titles · C++ | Unreal Engine 4/5

- Developed core kart physics and gameplay systems for DreamWorks All-Star Kart Racing, shipped across PC and console platforms.
- Implemented multiplayer kart replication with client prediction and interpolation supporting up to 8 players.
- Created design and automation tools to optimise workflow for game design and sandbox-oriented purposes, resulting in faster development cycles across multiple projects for a team of 4 designers, automating 2 games simultaneously.
- Improved the game engine with a specialised module, API, and tools for kart physics and mechanics, resulting in a 20% reduction in development time.

- Developed utility and debugging tools for tracking user traces to inform level design, resulting in a 40% level design time reduction.
- Tuned AI behaviour for decisions and mechanics in single-player kart adversaries, resulting in a 20% difficulty increase.
- Shipped menu systems and gameplay for Nickelodeon Kart Racers 3: Slime Speedway.

Saint Augustin National University — On-site (Arequipa, Peru)

Volunteer Game Programmer & Instructor (C++) | Apr 2020 – Oct 2022

C++ · Unreal Engine 4 · Curly Engine

- Shipped two indie titles (“Potato Smash” and a Volcano Simulation) as volunteer lead developer, designing and implementing features using C++ with Curly Engine and HSGIL.
- Contributed to multiple internal projects focused on gamification and educational game development, implementing gameplay systems and core mechanics.
- Developed and maintained game logic, systems, and interactive features across projects, alongside independent freelance development work during the same period.
- Mentored and trained groups of ~10 students in game development concepts, covering C++, custom engine tooling, and Unreal Engine 4.

SKILLS & INTERESTS

Technical: C++, C, C#, Rust, Python, Unreal Engine 4/5, OpenGL, DirectX 11/12, Vulkan, GLSL, Unity, Godot, Git, Perforce, Subversion.

Projects: Curly Engine & HSGIL — personal C++ game engine and graphics library, in continuous development since 2019; rendering foundation and engine used to ship indie titles for university classmates. github.com/AsulconS/Curly-Engine · github.com/AsulconS/HSGIL

Languages: Spanish (native); English (professional); Portuguese (conversational); Italian (intermediate).

Interests: Strategy board games (Go, Chess); MOBAs & RTS; science fiction & fantasy literature; acoustic music performance; hobbyist computational-science simulation projects.

EDUCATION

Universidad Nacional de San Agustín — Arequipa, Peru

Bachelor of Science in Computer Science | 2018 – Expected 2026

98% of degree requirements completed (220/224 credits).

Relevant Coursework: Data Structures, Algorithms, Software Engineering, Computer Graphics.